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MATERIAL SAFETY DATA SHEET

SECTION 1 CHEMICAL PRODUCTS & COMPANY IDENTIFICATION

CAS NUMBER: 7760-50-1
SUBSTANCE: BASIC MAGNESIUM CARBONATE
TRADE NAMES/SYNONYMS: MAGNESIUM CARBONATE GOLDSTAR / TWOSTAR / ONESTAR / P
MAGNESIUM CARBONATE HEAVY / LIGHT
TETRAKIS(CARBONATO(2-)DIHYDROXYPENTA-MAGNESIUM;
MAGNESIUM CARBONATE HYDROXIDE;
SYNTHETIC HYDROMAGNESITE; C4H2MG5O14 ; OHS73405
CHEMICAL FAMILY: Inorganic salt
CREATION DATE: 06/18/92 REVISION DATE: 01/05/99
CREATED DIVISION:

CHIEF OF INSPECTION SECTION

SECTION 2 COMPOSITION/INFORMATION ON INGREDIENTS

COMPONENT : BASIC MAGNESIUM CARBONATE
CAS NUMBER: 7760-50-1
PERCENTAGE: 100.0
OTHER CONTAMINANTS: NONE.

SECTION 3 HAZARDS IDENTIFICATION

CERCLA RATINGS (SCALE 0-3): HEALTH=U FIRE=0 REACTIVITY=0 PERSISTENCE=0
NFPA RATINGS (SCALE 0-4): HEALTH=1 FIRE=0 REACTIVITY=0
EMERGENCY OVERVIEW:
Odorless white powder.
Avoid breathing dust. Avoid contact with eyes, skin and clothing. Keep container tightly closed. Wash thoroughly after handling. Use only with adequate ventilation.
POTENTIAL HEALTH EFFECTS:
INHALATION:
SHORT TERM EFFECTS: May cause irritation.
LONG TERM EFFECTS: No information available on significant adverse effects.
SKIN CONTACT:
SHORT TERM EFFECTS: No information available on significant adverse effects.
LONG TERM EFFECTS: No information available on significant adverse effects.
EYE CONTACT:
SHORT TERM EFFECTS: May cause irritation.
LONG TERM EFFECTS: Same effects as short term exposure.
INGESTION:
SHORT TERM EFFECTS: May cause diarrhea and stomach pain.
LONG TERM EFFECTS: No information available on significant adverse effects.
CARCINOGEN STATUS:
OSHA: N
NTP: N
IARC: N

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SECTION 4 FIRST AID MEASURES

INHALATION:

FIRST AID- Remove from exposure area to fresh air immediately. Perform artificial respiration if necessary. Keep person warm and at rest. Treat symptomatically and supportively. Get medical attention immediately.

SKIN CONTACT:

FIRST AID- Remove contaminated clothing and shoes immediately. Wash with soap or mild detergent and large amounts of water until no evidence of chemical remains (at least 15-20 minutes). Get medical attention immediately.

EYE CONTACT:

FIRST AID- Wash eyes immediately with large amounts of water or normal saline, occasionally lifting upper and lower lids, until no evidence of chemical remains (at least 15-20 minutes). Get medical attention immediately.

INGESTION:

FIRST AID- If vomiting occurs, keep head lower than hips to help prevent aspiration. Treat symptomatically and supportively. Get medical attention if needed.

NOTE TO PHYSICIAN

ANTIDOTE: The following antidote has been recommended. However, the decision as to whether the severity of poisoning requires administration of any antidote and actual dose required should be made by qualified medical personnel.

MAGNESIUM COMPOUND POISONING:

When the serum magnesium level exceeds 5 meq/L and the patient exhibits symptoms, give calcium gluconate, 10 mL of 10% solution (intravenously) over several minutes. The dose may be repeated once; then the level should be checked again (Ellenhorn and Barceloux, Medical Toxicology, 1st Ed.).

Antidote should be administered by qualified medical personnel.

SECTION 5 FIRE FIGHTING MEASURES

FIRE AND EXPLOSION HAZARD:

Negligible fire hazard when exposed to heat or flame.

EXTINGUISHING MEDIA:

Dry chemical, carbon dioxide, water spray or regular foam (1993 Emergency Response Guidebook, RSPA P 5800.6).

For larger fires, use water spray, fog or regular foam (1993 Emergency Response Guidebook, RSPA P 5800.6).

FIREFIGHTING:

Move container from fire area if you can do it without risk. Do not scatter spilled material with high-pressure water streams. Dike fire-control water for later disposal (1993 Emergency Response Guidebook, RSPA P 5800.6, Guide Page 31).

Use agents suitable for type of surrounding fire. Avoid breathing hazardous vapors, keep upwind.

HAZARDOUS COMBUSTION PRODUCTS:

Thermal decomposition may release toxic oxides of carbon and magnesium.

SECTION 6 ACCIDENTIAL RELEASE MEASURES

OCCUPATIONAL SPILL:

Sweep up and place in suitable clean, dry containers for reclamation or later

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SECTION 7 HANDLING AND STORAGE

Observe all federal, state and local regulations when storing this substance.
Keep in a tightly closed container. Store in a cool, dry, ventilated area.
Store away from incompatible substances.

SECTION 8 EXPOSURE CONTROLS/PERSONAL PROTECTION

EXPOSURE LIMITS:

No occupational exposure limits established by OSHA, ACGIH, or NIOSH.

VENTILATION:

Provide local exhaust ventilation system.

EYE PROTECTION:

Employee must wear splash-proof or dust-resistant safety goggles to prevent eye contact with this substance.

Emergency eye wash: Where there is any possibility that an employee's eyes may be exposed to this substance, the employer should provide an eye wash fountain within the immediate work area for emergency use.

CLOTHING:

Employee must wear appropriate protective (impervious) clothing and equipment to prevent repeated or prolonged skin contact with this substance.

GLOVES:

Employee must wear appropriate protective gloves to prevent contact with this substance.

RESPIRATOR:

The following respirators are recommended based on information found in the physical data, toxicity and health effects sections. They are ranked in order from minimum to maximum respiratory protection.

The specific respirator selected must be based on contamination levels found in the work place, must be based on the specific operation, must not exceed the working limits of the respirator and must be jointly approved by the National Institute for Occupational Safety and Health and the Mine Safety and Health Administration (NIOSH-MSHA).

Any dust and mist respirator.
Any air-purifying respirator with a high-efficiency particulate filter.
Any powered air-purifying respirator with a dust and mist filter.
Any powered air-purifying respirator with a high-efficiency particulate filter.
Any type 'C' supplied-air respirator operated in the pressure-demand or other positive pressure or continuous-flow mode.
Any self-contained breathing apparatus.

FOR FIREFIGHTING AND OTHER IMMEDIATELY DANGEROUS TO LIFE OR HEALTH CONDITIONS:

Any self-contained breathing apparatus that has a full facepiece and is operated in a pressure-demand or other positive-pressure mode.

Any supplied-air respirator that has a full facepiece and is operated in a pressure-demand or other positive-pressure mode in combination with an auxiliary self-contained breathing apparatus operated in pressure-demand or other positive-pressure mode.

SECTION 9 PHYSICAL AND CHEMICAL PROPERTIES

DESCRIPTION:	Odorless white powder.
MOLECULAR FORMULA:	x(MgCO3) y(Mg(OH)2) z(H2O) (x:y:z= about 4:1:4)
MELTING POINT:	482 F (250 C) (decomposes)

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SPECIFIC GRAVITY:	2.16
WATER SOLUBILITY:	insoluble
VOLATILITY:	2.0-3.0%
EVAPORATION RATE:	(butyl acetate=1) nil
SOLVENT SOLUBILITY:	Soluble in dilute acids and ammonium salt solutions; insoluble in alcohol.

SECTION 10 STABILITY AND REACTIVITY

REACTIVITY:

Stable under normal temperatures and pressures.

CONDITIONS TO AVOID:

May burn but does not ignite readily. Avoid contact with strong oxidizers, excessive heat, sparks, or open flame.

INCOMPATIBILITIES:

ACIDS: Temperature and pressure increase in a closed container.

FORMALDEHYDE: Pressure increase in a closed container.

HAZARDOUS DECOMPOSITION:

Thermal decomposition may release toxic oxides of carbon and magnesium.

POLYMERIZATION:

Hazardous polymerization has not been reported to occur under normal temperatures and pressures.

SECTION 11 TOXICOLOGY INFORMATION

CARCINOGEN STATUS:	None.
ACUTE TOXICITY LEVEL:	No data available.
TARGET EFFECTS:	No data available.

AT INCREASED RISK FROM EXPOSURE: Persons with kidney impairment.

** May be based on general magnesium salt information.

HEALTH EFFECTS

INHALATION:

ACUTE EXPOSURE- May cause irritation of the respiratory system.

CHRONIC EXPOSURE- Repeated or prolonged exposure to dust may cause bronchitis.

SKIN CONTACT:

ACUTE EXPOSURE- Essentially no irritation.

CHRONIC EXPOSURE- Repeated or prolonged contact may result in allergic reactions.

EYE CONTACT:

ACUTE EXPOSURE- May cause irritation.

CHRONIC EXPOSURE- Repeated or prolonged exposure to magnesium salts may cause conjunctivitis.

INGESTION:

ACUTE EXPOSURE- Ingestion may cause allergic reactions. Doses of 600 mg produces an antacid effect. Large amounts of magnesium salts may cause diarrhea and abdominal pain. More serious symptoms of hypermagnesemia, such as electrolyte imbalance, central nervous system depression with a drop in blood pressure and respiratory paralysis, and neurological and cardiac impairment, are rare in the absence of intestinal or renal disease since magnesium is poorly absorbed from the gastrointestinal tract and readily excreted by the kidneys.

CHRONIC EXPOSURE- May cause iron deficiency. Magnesium preparations may cause phosphorus depletion syndrome.

SECTION 12

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ENVIRONMENTAL IMPACT RATING (0-4):	no data available
ACUTE AQUATIC TOXICITY:	no data available
DEGRADABILITY:	no data available
LOG BIOCONCENTRATION FACTOR (BCF):	no data available
LOG OCTANOL/WATER PARTITION COEFFICIENT:	no data available

SECTION 13 DISPOSAL INFORMATION

Observe all federal, state and local regulations when disposing of this substance.

SECTION 14 TRANSPORTATION INFORMATION

No classification currently assigned.

SECTION 15 REGULATORY INFORMATION

TSCA STATUS:	Y
CERCLA SECTION 103 (40CFR302.4):	N
SARA SECTION 302 (40CFR355.30):	N
SARA SECTION 304 (40CFR355.40):	N
SARA SECTION 313 (40CFR372.65):	N
OSHA PROCESS SAFETY (29CFR1910.119):	N
CALIFORNIA PROPOSITION 65:	N
SARA HAZARD CATEGORIES, SARA SECTIONS 311/312 (40 CFR 370.21)	
ACUTE HAZARD:	N
CHRONIC HAZARD:	N
FIRE HAZARD:	N
REACTIVITY HAZARD:	N
SUDDEN RELEASE HAZARD:	N

SECTION 16 OTHER